Session 8 — DAX Measures and Filter Context

Milestone B5 · Asynchronous activity · BDAT04 Fundamentals of Data Warehousing · Term 1, AY 2026–2027

RELEASED	Thursday, 15 October 2026
DUE	**12:00 noon, Friday 16 October 2026** — the §IX-C late cap applies after this.
SUBMIT TO	`b5-dax-measures/` in your folder of the class repository
MARKED ON	Rubric D (printed on the last page), out of 100
PREREQUISITE	Your finished B4 model. **If your relationships or dimDate are not right, nothing in this activity will work.**
PLATFORM	Power BI Service in the browser, My workspace. Model view and DAX query view.
EFFORT	About three hours. Six measures and one calculated column — the writing is the short part.

The AI rule for this activity — read it twice

DAX written by an assistant is Tier 3 and needs my written approval in advance. That includes Copilot. If the Fabric trial is active on your account, Copilot will offer to write measures for you. **Decline it.** For this activity the answer is no — there is no approval on file, and a measure you did not write is a measure you cannot defend in the oral.

What is allowed (Tier 2, log it): asking an assistant to explain what CALCULATE does, or to explain an error message you are already looking at. Asking it what a function means is learning. Asking it for the expression is not.

If Copilot writes something before you can stop it, do not delete the evidence. Say so in your AI Use Log, say what you did about it, and keep going. An honest log is never penalized. This one will be read carefully.

What you are building, and why it matters

In B4 you built the model. It holds no answers yet — a star schema with no measures is a filing cabinet. This week you write the measures that turn it into something a branch manager can ask questions of.

There is one idea underneath everything in this activity, and if you leave with only this, the week has done its job: **a measure has no value of its own.** It has a value *in a context*. The same expression gives you a whole year of company revenue in one cell and a single branch's worst Tuesday in the cell beside it, and nothing about the expression changed. Part B is three items long because this is the idea students most often carry a wrong version of into their capstone.

You are not writing time intelligence this week — no year-to-date, no same-period-last-year. That is Week 10, and it will rest on the dimDate you built in B4.

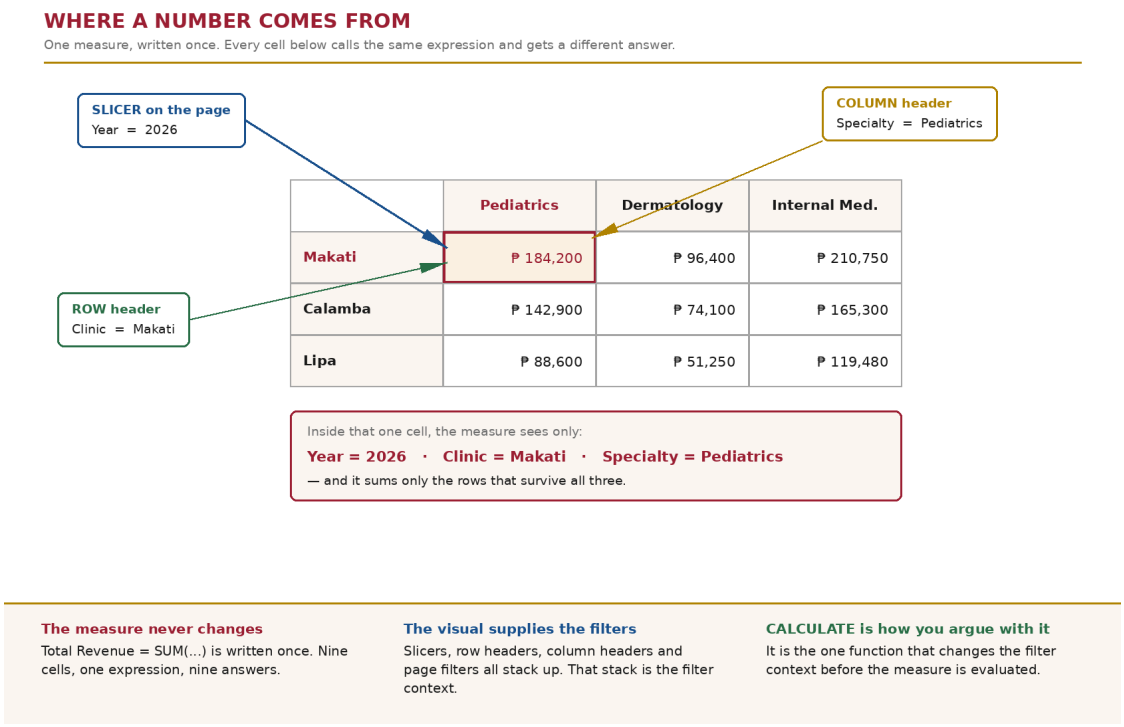


Figure 1. Where a number comes from. Drawn on a clinic example so that it teaches the reading rather than handing you your own matrix.

Part A — Measures, and what makes them different (about 50 minutes)

ITEM 1 Write the three base measures, then prove one of them 30 min

Create these as **measures**, not calculated columns. Put them in the sales table.

```
Total Revenue      = SUM ( tblSales[Amount] )
Total Units        = SUM ( tblSales[Quantity] )
Transaction Count  = DISTINCTCOUNT ( tblSales[SaleID] )
```

1.1 Put Total Revenue on a card visual with no filters anywhere on the page.

1.2 Compare that number against the known total you recorded in **B4, Item 7.1**. They must match exactly. If they do not, stop and find out why before going further — something in your model changed.

1.3 Now put `Transaction Count` beside a second measure defined as `COUNTROWS ( tblSales )`. Record both numbers. Are they the same, or different?

ANALYSIS — WHAT YOU FOUND

Total Revenue is 2,694,600.80, which matches the known total from B4. Transaction Count is 11,167 and Row Count is also 11,167, so the two counts are the same.

Reflection prompt. Explain what you found, using the grain statement you wrote in B4 Item 2. Then think back to Week 4, when you removed 23 duplicate SaleIDs: what would these two measures have shown if you had *not* removed them? Finish by saying what that makes this pair of measures useful for, every time new data lands.

REFLECTION — WHAT IT MEANS

One row in tblSales represents one sale, so the row count and distinct SaleID count should match. If the 23 duplicate SaleIDs from Week 4 had not been removed, the row count would be higher than the transaction count. Comparing these two measures is useful for checking new data for duplicate sales.

ITEM 2 Safe division, and what a manager sees when it is not 20 min

2.1 Create **`Average Transaction Value`** using DIVIDE:

```
Average Transaction Value = DIVIDE ( [Total Revenue], [Transaction Count] )
```

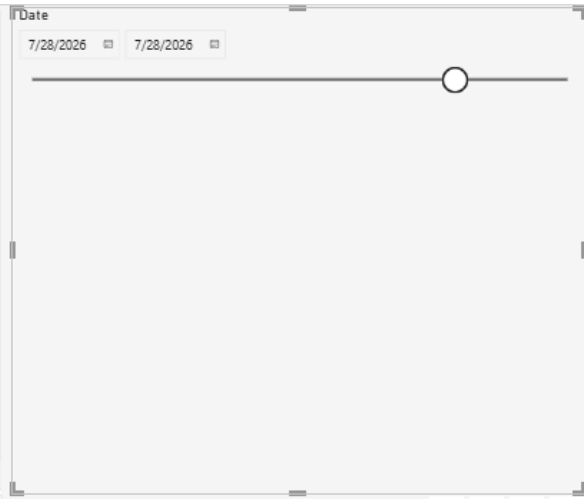
2.2 Now create a second, deliberately wrong version using the division operator:

```
ATV Unsafe = [Total Revenue] / [Transaction Count]
```

2.3 Put both in a table by Branch, then filter the page to a date on which one branch sold nothing. Record what each measure returns in that row.

2.4 Delete ATV Unsafe when you are done. Keep the screenshot.

ATV Unsafe	Average Transaction Value	BranchName
182.50	182.50	
312.86	312.86	Binan
211.00	211.00	Cabuyao
396.67	396.67	Calamba
		Los Banos
350.00	350.00	San Pablo
235.00	235.00	Santa Rosa
298.33	**298.33**	



ANALYSIS — WHAT YOU FOUND

On 7/28/2026, Los Banos had no sales. Both Average Transaction Value and ATV Unsafe returned a blank value.

Reflection prompt. One of those two results appears on a branch scorecard in front of the owner. Describe what she would conclude from it, and say why that is worse than a blank cell.

REFLECTION — WHAT IT MEANS
A blank value tells the owner that there were no transactions to calculate an average from. Showing a misleading number would be worse because it could make the owner think the branch had actual sales activity when it did not.

Part B — Filter context (about 80 minutes)

This is the part of the activity that carries the most marks. Work slowly. Every item here is a different way of asking the same question: *what can this measure see?*

ITEM 3 One measure, nine answers 25 min

3.1 Build a matrix visual: **Branch** on rows, **Category** on columns, `Total Revenue` in values.

3.2 Add a slicer for Customer Tier and leave it unselected for now.

3.3 Pick one cell — any cell that is not a total. Write down its exact value, its row, and its column.

3.4 Now list *every* filter acting on that one cell. There are more than two. Include anything coming from the visual itself, from the page, and from the relationships you built in B4.

3.5 Screenshot the matrix. Name it matrix.png — it is part of your submission.

ANALYSIS — WHAT YOU FOUND

I selected the Binan–Merchandise cell, which has a Total Revenue of 1,490. The filters acting on this cell are BranchName = Binan from the matrix row and Category = Merchandise from the matrix column. The Customer Tier slicer is unselected, so all customer tiers are included. There are no additional page filters applied. The relationships from tblBranches and tblProducts to tblSales pass the branch and category filters into the sales table, which determines the revenue shown in the cell.

Reflection prompt. You wrote Total Revenue once. It appears in that matrix a dozen times and returns a different number nearly every time, without being edited. Explain, in your own words, where the difference comes from.

REFLECTION — WHAT IT MEANS

I only created Total Revenue once, but each matrix cell gives a different answer because each cell has a different filter context. The branch row and product category column filter the sales data differently, so the same measure calculates revenue only for the records that match that specific combination.

ITEM 4 Argue with the filter context 30 min

CALCULATE is the only function that changes the filter context before a measure is evaluated. This item is the smallest honest demonstration of it.

4.1 Create these two measures:

```
Revenue All Branches   = CALCULATE ( [Total Revenue], REMOVEFILTERS ( tblBranches ) )

Branch Share of Revenue = DIVIDE ( [Total Revenue], [Revenue All Branches] )
```

4.2 Put Branch on rows with all three measures. Format the share as a percentage.

4.3 Does the share column add up to 100%? Record the actual total, not the one you expect.

4.4 Now set the Customer Tier slicer to a single tier. Watch `Revenue All Branches` carefully and record whether it changes.

ANALYSIS — WHAT YOU FOUND

With no Customer Tier selected, Revenue All Branches was 2,694,600.80. The Branch Share of Revenue total was 100.0%. When I selected Gold, Revenue All Branches changed to 102,125.19, so the measure still responds to the Customer Tier filter.

Reflection prompt. That measure is called “all branches”, and yet it moved when you touched a slicer that has nothing to do with branches. Explain why — precisely, naming what REMOVEFILTERS did and did not do. Then say what you would have to change if you wanted a true grand total that ignores everything.

REFLECTION — WHAT IT MEANS

REMOVEFILTERS(tblBranches) only removes the branch filter. It does not remove the Customer Tier filter, so Revenue All Branches changed when I selected Gold. If I wanted a true grand total that ignored everything, I would also need to remove the other filters affecting the measure.

ITEM 5 When a calculated column is the right answer 25 min

You have been told measures are usually right. “Usually” is doing work in that sentence. This item is where you find the line.

5.1 Create a **calculated column** in the sales table:

```
LineCost = tblSales[Quantity] * RELATED ( tblProducts[UnitCost] )
```

5.2 Then create two measures:

```
Total Cost     = SUM ( tblSales[LineCost] )
```

```
Gross Margin % = DIVIDE ( [Total Revenue] - [Total Cost], [Total Revenue] )
```

5.3 Now do the wrong thing on purpose. Create a **calculated column** as well:

```
Margin Pct Column =
    DIVIDE ( tblSales[Amount] - tblSales[LineCost], tblSales[Amount] )
```

5.4 Put Branch on rows. Add `Gross Margin %` and the average of `Margin Pct Column`. Record both numbers for every branch.

5.5 Delete Margin Pct Column when you have your screenshot. Keep LineCost — it is correct and you will use it again.

ANALYSIS — WHAT YOU FOUND

Gross Margin % and Average of Margin Pct Column gave different results for every branch. Binan was 59.27% vs 60.65%, Cabuyao 58.21% vs 60.27%, Calamba 58.62% vs 60.48%, Los Banos 58.67% vs 60.53%, San Pablo 58.69% vs 60.41%, and Santa Rosa 58.80% vs 60.58%. Overall, Gross Margin % was 58.72%, while the average of Margin Pct Column was 60.50%.

Reflection prompt. LineCost is a column and it is right. Margin Pct Column is a column and it is wrong. State the rule that separates them in one sentence, then apply your rule to a case I have not given you: should “days since the customer joined” be a column or a measure?

REFLECTION — WHAT IT MEANS

A calculated column is correct when the value belongs to each individual row, while a measure is better when the result depends on totals or filters. LineCost belongs to each sale, so a column is correct. Margin percentage should be a measure because averaging row percentages gives a different result. “Days since the customer joined” should be a measure because it changes as time passes.

Part C — Make it survive contact (about 50 minutes)

ITEM 6 Name it, format it, explain it 20 min

A measure nobody can read is a measure nobody will trust. This is the cheapest mark on the sheet and the one most often skipped.

6.1 Put every measure in a display folder. Delete any stray measure you created while experimenting.

6.2 Set the format on each one: peso and no decimals for money, one decimal and a percent sign for rates, thousands separators throughout.

6.3 Write a one-line description on each measure, in the model, in business language. Not “sums the Amount column” — say what the number *is*.

6.4 Record all of it in a table: measure name, what it answers, format applied.

ANALYSIS — WHAT YOU FOUND

Measure name	What it answers	Format applied
Total Revenue	Total sales revenue generated by the business	₱, 0 decimals
Total Units	Total number of units sold	Whole number
Transaction Count	Number of unique sales transactions	Whole number
Average Transaction Value	Average revenue per sales transaction	₱, 0 decimals
Revenue All Branches	Total revenue across all branches within the remaining filters	₱, 0 decimals
Branch Share of Revenue	Current branch’s share of total branch revenue	Percentage, 1 decimal
Total Cost	Total cost of products sold	₱, 0 decimals
Gross Margin %	Percentage of revenue remaining after product cost	Percentage, 1 decimal

Reflection prompt. Name one measure whose name you changed while doing this, and say what was dangerous about the first name. If you changed none, say why your first names were already safe — and be honest.

REFLECTION — WHAT IT MEANS

I did not change any measure names because the original names were already clear and described what each measure represented. Changing them was not necessary and could make the model less consistent.

ITEM 7 Break them on purpose 30 min

Rubric B's “Developing” level describes measures that “return correct totals but break under filtering.” This item is how you find out which of yours do.

7.1 Filter the page to a branch and a category that branch has never sold. Record what every measure returns.

7.2 Filter to a date outside the range of your data. Record every measure again.

7.3 Filter to the blank customer — the one you found on the dimension side in B4 Item 7. Record every measure a third time.

7.4 Build a small grid of your results: three scenarios down, your measures across.

ANALYSIS — WHAT YOU FOUND

Scenario	Total Revenue	Total Units	Transaction Count	Avg. Transaction Value	Revenue All Branches	Branch Share	Total Cost	Gross Margin %
Branch + category never sold	**No such combination found**	—	—	—	—	—	—	—
Date outside sales range: 8/1/2026	Blank	Blank	Blank	Blank	Blank	Blank	Blank	Blank
Blank customer	₱1,226,559.26	7,769	5,047	₱243	₱1,226,559	100.0%	₱506,571	58.7%

Reflection prompt. Which measure behaved worst, and what did it return? Then answer the question that actually matters: when there is genuinely nothing to count, is a blank cell or a zero the more honest thing to show a manager? Pick one and defend it — there is a defensible case for either, and I am marking the defense, not the choice.

REFLECTION — WHAT IT MEANS

The measures did not show an unexpected failure, but the no-data date scenario made every measure return blank. I think blank is more honest than zero because zero could suggest that a real transaction or calculation occurred and resulted in zero, while blank shows that there was no data available to calculate from. The blank customer test also showed that unmatched customer records are grouped under the blank member rather than disappearing from the model.

What to submit

Everything into **`b5-dax-measures/`** in your own folder of the class repository, by **12:00 noon on Friday 16 October**.

- **`B5_DAX_Measures_<SURNAME>.docx`** — this sheet, with every Analysis and Reflection box filled in.
- **`model_2026-10-16.tmdl`** — the TMDL export of the finished model, in your backups folder. **It contains every measure expression you wrote.** I will read it against your sheet.
- **`matrix.png`** — the Item 3 matrix, with the slicer visible.
- **`ai-use-log.md`** — updated for this milestone. If Copilot appeared and you declined it, that is a log entry and a good one.

> ### If you get stuck
>
> Post in the group chat, not a private message. DAX errors are almost always one of four things, and whichever one is biting you is probably biting someone else.
>
> Before you ask, check: is the measure in the right table, are the square brackets and quote marks right, does the column you referenced still exist under that name, and is your B4 relationship actually there? That last one accounts for more "my measure returns blank" messages than the other three combined.

How this is marked — Rubric D

The course rubric for build milestones B3 to B7. Each criterion is scored on a four-point scale and weighted; your mark is the sum of (level × weight) ÷ 4.

CRITERION	WEIGHT	WHAT I AM LOOKING FOR
Technical correctness	**35**	Measures return correct values and hold up under filtering. DIVIDE used. The column-versus-measure decision in Item 5 is right and is justified.
Business justification	**25**	Every measure answers a business question you can state. Filter-context explanations are in your own words, not a definition copied from somewhere.
Documentation and readability	**20**	Measures named, formatted, foldered and described in the model. No stray experiments left behind. matrix.png present.
Response to prior feedback	**15**	What your B4 letter asked for, visibly acted on. The TMDL export is checked here — it carries every measure you wrote, verbatim.
Timeliness and integrity	**5**	In the right folder by noon. AI Use Log complete and specific. The DAX in your model is DAX you wrote.

Lagdamen, Lawrence M.
NAME AND SIGNATURE

10/14/2026
DATE AND TIME SUBMITTED